

Tests & Diagnostics — AGEL0206 σ_e ApJL paper

Last updated: 2026-05-27 (M11 rigorous re-derivation of carried systematics)

Headline (NEW `_mtwdo_` + bad-pixel mask + Balmer-unmasked + He I 3819 mask + M10 full sky-line audit + M11 cube-matched systematic re-derivation + M12 `R_e`-source fold-in, wide arc-masked window): $\sigma_e(<R_e) = 270 \pm 13$ km/s = 269.62 ± 13.27 km/s (stat $\pm 5 \oplus$ I-shape $\pm 2.27 \oplus$ mask $\pm 6.65 \oplus$ frame $\pm 5 \oplus$ centering $\pm 4 \oplus$ window $\pm 3.82 \oplus$ reduction $\pm 3.45 \oplus$ `R_e`-source ± 6.13) Asymmetric form: $269.62 -13.45 / +13.10$ km/s. All headline numbers are emitted deterministically by `scripts/paper_values.py` $\rightarrow$ `results/PAPER_VALUES.json` (single source of truth; do not hand-edit). Source: `scripts/run_wide_sigma_e.py --cube new_clean_hei`. M11 sweep caches: `results/{ishape_sweep,maskweight_sweep}_wR3800_5400_arcmask_M10/` + `results/nb09a_wavelength_sweep_M10/`. The carried ± 15 fit-window component dropped to ± 3.82 on the cleaned NEW cube — the dominant source of the ± 6 km/s total-budget tightening.

Second-reduction cross-check (OLD cube + bad-pixel mask + Balmer-unmasked + He I + M10 sky masks): $\sigma_e(<R_e) = 262.72 \pm 17.99$ km/s (asymmetric $-18.10 / +17.88$). The 6.90 km/s Δ between the two CLEANED + He-I + M10-sky-masked reductions sets the refined ± 3.45 km/s reduction-pass systematic (was ± 3.86 from clean+He-I-only cubes; M10 sky masks tighten the inter-reduction gap further).

Narrow-window cross-check (Ca H+K + G, OLD cube): $\sigma_e(<R_e) = 268 \pm 30$ km/s = 267.95 ± 30.10 km/s (stat $\pm 24 \oplus$ I-shape $\pm 3.7 \oplus$ mask $\pm 7.5 \oplus$ frame $\pm 5 \oplus$ centering $\pm 4 \oplus$ window ± 15)

Three-way consistency: the headline (271.87, new clean + He I) and narrow-window cross-check (267.95, old narrow) are within 4 km/s — both inside each other's 1σ stat error. The pre-He-I new-cube wide-window value (268.98) was within 1 km/s of the narrow window; adding the He I mask moves the wide headline ~ 3 km/s redder of the narrow value, but still well within consistency.

Method (wide, headline): Single ppxf fit on the I-weighted aperture spectrum at `R<R_e=2.305` over rest 3800–5400 Å with explicit `z=1.302` source-emission masks (MgII $\lambda 2800$, [OII] $\lambda 3727$, He I 3819 (M8), [NeIII]/Mg b cluster) + bad-pixel mask + Balmer kept in fit, F200LP spatial arc mask, frame-aware (per-SPS vac/air), SPS-pooled (FSPS+EMILES+XSL) wild bootstrap $\times$ polynomial-degree posterior.

Source-emission verification figure (appendix candidate): integrated arc spectrum at AGEL_0206_ApJL_Figures/AGEL0206_arc_source_spectrum.pdf — direct extraction from the 69 arc-flagged spaxels (deflector continuum subtracted) showing [OII] 3727 + H9 + [NeIII] + He I 3819 in source-rest frame and the same lines mapped to def-rest (the headline fit window).

Method (narrow, cross-check): Same machinery but on obs-frame 6500–7500 Å (rest 3879–4476 Å, anchored on Ca H+K and G-band only) — the §6cum cumulative I-weighted ppxf path retained from nb07c.

This document is the canonical index of every test, sweep, audit, and sensitivity check run for the σ_e measurement. Each row points to the script/notebook that runs it and the result file or note it produced.

0. Quick-reference test matrix

#	Test	Where	Result	Status
A. Foundational				

#	Test	Where	Result	Status
A1	KCWI cube provenance — Keck observing logs cross-checked	NOTES_methodology_2026-04-27.md, reference_kcwi_data_properties.md (updated 2026-05-26)	Verified contributing nights: K409 UT 2025 Aug 30 (12 RED + 4 BLUE DESJ0206 frames) + U002 PI Jones UT 2025 Nov 17 (20 RED + 5 BLUE DESJ0206 frames). Dec 29 2024 RED frames flagged pending FITS-header verification. Sept 29 K409 has zero DESJ0206 entries — earlier note was wrong.	✓
A2	Systemic redshift via line fitting	notebooks/04_redshift_verification.ipynb, scripts/redshift_verify.py	$z = 0.67564$ (vs DR2 0.67511)	✓
A3	R_e from masked curve-of-growth (F140W+F200LP)	scripts/measure_Re.py, scripts/final_sigma_e.py:curve_of_growth	$R_e = 2.305'' = 16.23$ kpc	✓
A4	HST-mean center via centroid_2dg	scripts/final_sigma_e.py:find_center	F140W & F200LP agree to 0.36''	✓
A5	Bagpipes SED fitting ($M_\star$) — aperture vs Sersic-total	notebooks/02_streamlined_Bagpipes_SED.ipynb, 08_sersic_total_photometry.ipynb	$\log M_\star = 11.33$ +0.07/−0.09 (old expert-aperture); 11.40 +0.11/−0.15 (Sersic-total). Superseded by N-series principled masking → headline 11.16 +0.31(sys)	✓
B. Pipeline correctness audits				
B1	Instrumental LSF audit (DISPSCAL=0.294)	scripts/ifu_spectral_resolution.py	FWHM=0.692 Å; σ_v inst $\approx$ 12-14 km/s	✓
B2	σ_{inst} sensitivity ($\times 0.5$ to $\times 2.0$ LSF)	scripts/sigma_inst_sensitivity.py, audit 3	$\max \Delta\sigma = 0.83$ km/s	✓
B3	SPS template wavelength frames	scripts/audit_ppxf_methodology.py, NOTES Test 2	FSPPS=vacuum, EMILES=air, XSL=air (Ca K minimum + V_{sys})	✓
B4	V_{sys} air vs vacuum $\times 5$ polynomial degrees	scripts/audit_ppxf_methodology.py audit 1	ΔV consistent within ± 2 km/s across degrees	✓
B5	$z \times$ air-vac differential (obs vs rest application)	scripts/audit_ppxf_methodology.py audit 2	1.83 km/s differential — sub-budget	✓
B6	fw_hm_gal_dict frame consistency check	scripts/audit_ppxf_methodology.py audit 4	dict in REST frame, matches Cappellari pattern	✓
C. SPS templates & frame-fix				
C1	Frame-aware ppxf: per-SPS native frame	scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME + frame_galaxy='auto'	V_{sys} split collapsed from ~ 110 to ~ 15 km/s	✓
C2	End-to-end V_{sys} closure (frame swap)	NOTES Test 3, addendum 2026-04-29	Frame fix matches diagnosis	✓
C3	σ shift from frame fix	NOTES addendum, error_budget()	≤ 5 km/s across SPS → carried as ± 5 km/s budget	✓
C4	3-SPS pooled posterior	scripts/final_sigma_e.py:pool_sps	Per-SPS V_{sys} subtracted before pooling	✓
D. Aperture & $I(r)$				
D1	$\S 6$ cumulative I-weighted ppxf	scripts/final_sigma_e.py:run_aperture_sps	$\sigma_e(<R_e) = 267.82$ km/s headline	✓

#	Test	Where	Result	Status
D2	§6cum vs §7 (annular) cross-check	notebooks/07c, 7.claude/.../reference_cumulative_vs_annular_sigma_e.md	§7=255, §7b=271 — all <1 σ from §6cum	✓
D3	I(r) shape sweep (10 shapes, fixed mask)	scripts/run_isource_shape_sweep.py, scripts/analyze_isource_shape_sweep.py	Range = 12.9 km/s (excluding F200LP_Sersic2D outlier) → ± 13 budget	✓
D4	I(r) shape sweep refresh post-frame-fix at N=500	NOTES addendum (d), 2026-04-29	Frame fix has <0.5 km/s impact; ± 13 budget validated	✓
D5	Aperture choice: $R < R_e/8$, $R < R_e/2$, $R < R_e$	scripts/final_sigma_e.py: APERTURE_FRACS	$R < R_e/8$ dropped (inside seeing FWHM/2 = 0.64")	✓
D6	Equal-N vs equal-width annular binning	notebooks/07b (5-bin), notebooks/07c (equal-N)	Equal-N inside $R_{\text{safe}} = 3R_e/4$ chosen	✓
D7	R_e source systematic (mean vs F140W vs F200LP vs CaHK+G)	scripts/final_sigma_e.py §7, notebooks/09 §7b (narrow); scripts/run_sigma_e_Re_systematic_wide.py, nb15 (wide)	NARROW: spread = 16.9 km/s (mean=268, F140W=265, F200LP=276, CaHK=282). WIDE (2026-06-08, N=500): spread = 12.25 → ± 6.13 km/s (mean 2.305"=269.62, F140W 2.168"=267.44, F200LP 2.441"=272.44, CaHK+G 2.902"=279.69). FOLDED into the M12 budget (was sub-budget only). Light- R_e -family-only spread = ± 2.50 (cross-check).	✓ folded (M12)
E. Mask treatment				
E1	F200LP arc mask reprojected to IFU grid	scripts/final_sigma_e.py: load_setup	0.7% of all spaxels flagged (~38 inside $R < R_e$)	✓
E2	Hard mask (w=0.0) headline	scripts/final_sigma_e.py, track A	$\sigma_e = 267.82 \pm 24$ km/s	✓
E3	No-mask sensitivity (w=1.0)	track B	$\sigma_e = 252.82$, $\Delta = -15.0$ km/s	✓
E4	Soft-mask single-point (w=0.5)	scripts/soft_mask_track.py, track C, NOTES addendum (b)	$\sigma_e = 258.54$, $\Delta = -9.29$; super-linear	✓
E5	5-point mask-weight sweep $w \in \{0, 0.25, 0.5, 0.75, 1\}$	scripts/soft_mask_track.py --weight ..., scripts/analyze_mask_weight_sweep.py, NOTES addendum (c)	Per-step drops 5.0 → 4.3 → 3.5 → 2.2; quadratic c=+7.3 (concave-up); threshold-dominated	✓
E6	Mask dilation (over-masking diagnostic)	notebooks/07d_sigma_e_forceful_mask.ipynb, 7.claude/.../project_nb07d_overmasking_finding.md	Dilation inflates σ via noise; do NOT dilate (negative finding)	✓
E7	Arc-spectrum subtraction sibling	notebooks/07e_sigma_e_arc_subtract.ipynb	Matches §6cum within 0.1 km/s — residual arc dilution sub-dominant	✓

#	Test	Where	Result	Status
E8	Arc-as-sky mechanism test (no-mask + arc-sky template)	scripts/run_07f_arc_sky.py, notebooks/07f_arc_sky_subtract.ipynb	$\sigma_D=274.6$ vs $\sigma_A=267.8$ / $\sigma_B=252.8$; recovery=145% — dilution explains the full no-mask Δ	✓
F. Centering				
F1	5-center sweep ($\pm 0.4''$ perturbations)	NOTES_centering_investigation_2026-04-27.md, scripts/regen_s6cum_nomask_diagnostic.py	σ_e spread = 3.7 km/s $\rightarrow$ ± 4 km/s budget	✓
F2	HST F140W vs F200LP centroid offset	scripts/final_sigma_e.py:find_center	0.36'' — driven by F200LP arc; HST-mean robust	✓
G. Bootstrap & polynomial degree				
G1	Wild bootstrap (Rademacher $\times$ local residual)	scripts/bootstrap_ppxf.py:run_bootstrap_single_degree	75-pix rolling window for σ_{loc}	✓
G2	Parallel bootstrap (joblib, BLAS=1 per worker)	scripts/bootstrap_ppxf_parallel.py	~6 min for 6 fits at N=500	✓
G3	Polynomial degree sweep (deg 15-29)	notebooks/03c, scripts/build_nb09.py:\$6.5	σ flat within bootstrap envelope; polynomial saturated	✓
G4	N=50 smoke vs N=500 production agreement	scripts/final_sigma_e.py --n_bootstrap	Within 1 km/s	✓
H. Cross-checks (independent estimators)				
H1	§7 discrete Gültekin annular sum (frame-aware, refresh 2026-05-01)	scripts/refresh_07c_gultekin.py, notebooks/07c §7	256.17 $-13.0/+12.7$ km/s (post-fix; was 254.99 $-24/+28$ pre-fix)	✓
H2	§7b flat- σ extrapolation (frame-aware, refresh 2026-05-01)	scripts/refresh_07c_gultekin.py, notebooks/07c §7b	274.37 $-16.2/+17.4$ km/s (post-fix; was 271 $-33/+35$ pre-fix)	✓
H3	F200 mask sensitivity at N=500	results/annular_bootstrap_07c_nomask/	$\Delta_{mask} = -16.4$ km/s $\rightarrow$ ± 16 budget	✓
I. Earlier sigma-discrepancy work (Tests 1-21)				
I1	NB01 vs NB05 σ reproduction	notebooks/05x Tests 1-4	Confirmed σ depends on aperture	✓
I2	Source mask + PSF-aware masking	notebooks/05x Tests 7-10	F140W mask flags deflector core; F200LP mask is the right arc mask	✓
I3	Threshold sweep + contamination weighting	notebooks/05x Tests 11-13	Pre-Gültekin masking strategy work	✓
I4	V_rms profile + bootstrap	notebooks/05x Tests 14-16	Pre-Gültekin radial work	✓
I5	Redshift sensitivity ($z=0.67511$ vs 0.67564)	notebooks/05x Test 17	<1 km/s effect on σ	✓
I6	Voronoi binning attempt	notebooks/05x Test 18	Failed — abandoned	✓
I7	PowerBin spatial binning	notebooks/05x Test 19, scripts/run_powerbin_test19.py	Rejected as binning scheme — outer-bin $\sigma > 800$ km/s	✓
I8	R_e four-way comparison	notebooks/05x Test 20, scripts/measure_Re.py	F140W+F200LP masked CoG mean = 2.305'' headline	✓
I9	Per-spaxel vs PowerBin rotation map	notebooks/05x Test 21	No rotation on Re-scale; σ -dominated	✓

#	Test	Where	Result	Status
J. Wide arc-masked window (2026-05-08 → 2026-05-13)				
J1	Wavelength-window sweep (15 windows, narrow → wide)	notebooks/09a_wavelength_window_sweep.ipynb, scripts/run_window_sweep.py	wR3800_5400 wins on stat precision; XSL template floor at def-rest 3675 Å sets blue edge	✓
J2	Discovery of $z=1.302$ source-emission contamination	notebooks/09a §3a, notebooks/09b §8	Bimodal ppxf posterior in wide window ($\sigma \approx 250$ vs 100–150) caused by source emission lines mapped to deflector rest frame	✓
J3	Spectral-mask catalog: 3 source-emission + 1 telluric, def-rest	scripts/run_window_sweep.py: ARC_MASKS_REST	Four bands (def-rest 3835–3855 = Mg II $z=1.302$; 4525–4545 = O ₂ A-band telluric edge at obs 7593–7626 Å; 5115–5135 = [O II] $z=1.302$; 5260–5340 = [Ne III] $z=1.302$); 212 of 2374 pixels (8.9%) masked. Telluric ID via NOTES_4534A_spike_investigation_2026-05-18.md	✓
J4	Arc-mask + ppxf clean=True diagnostic	notebooks/09b §8	Bimodality collapses; σ vs polynomial degree FLAT across deg 15–29; clean=True drops 0 pixels	✓
J5	Per-SPS posterior collapse at wide arc-masked	notebooks/09d §1.3	SPS spread (FSPS / EMILES / XSL): 26.0 km/s (narrow) → 4.2 km/s (wide arc-masked); SPS systematic essentially disappears	✓
J6	N=500 production at wide arc-masked window	results/nb09a_wavelength_sweep/wR3800_5400_arcmask*_N500.npz	$\sigma_e = 254.9 - 7/+5$ km/s (stat-only, all-3 SPS pool)	✓
J7	Sersic2D bound-fix ($n \in [1.0, 6.0]$)	scripts/run_isource_shape_sweep.py: 181–207, 2026-05-11	F200LP previously escaped to $n=0.30$ (unphysical flat-disk), inflating I-shape budget; multi-init grid + tight bounds fix it. I-shape $\pm 5.4 \rightarrow \pm 1.5$	✓
J8	I-shape sweep at wR3800_5400_arcmask (10 shapes $\times$ 3 SPS $\times$ N=250)	results/ishape_sweep_wR3800_5400_arcmask/	std = 1.5 km/s — 9 $\times$ tighter than narrow's ± 13	✓
J9	Mask-weight sweep at wR3800_5400_arcmask ($w \in \{0, 0.5, 1.0\} \times 3$ SPS $\times$ N=250)	results/maskweight_sweep_wR3800_5400_arcmask/	peak-to-peak/2 = 3.8 km/s — 4 $\times$ tighter than narrow's ± 16 ; spectral arc mask absorbs the worst	✓
J10	Three-window cross-check at N=500	nb09 §9: w6500_7500 vs wR3800_5400_arcmask vs wR4000_5400_arcmask	Spread = 15.0 km/s → window systematic ± 15 (dominant residual). wR4000_5400_arcmask = orthogonal H β + Mg b feature set (no Ca H+K)	✓

#	Test	Where	Result	Status
J11	Both-windows side-by-side budget	notebooks/09d_final_ systematics_both_ windows.ipynb, results/sigma_e_ final_systematics_ nb09d.npz	WIDE 254.85 ± 17.87 vs NARROW 267.95 ± 30.10 ; consistent at 0.4σ	✓
K. Resolved kinematics (now feasible with wide arc-masked window)				
K1	Per-spaxel ppxf inside $R < 1.5 R_e$ at $S/N \geq \{2,3,5,10\}$	notebooks/11_perbin_ perspaxel_ kinematics_wide. ipynb	$S/N \geq 5$ gives 17 spaxels, $\sigma \in [144, 251]$ km/s with 0 outliers — first clean per-spaxel σ map for this target	✓
K2	PowerBin spatial binning at wide arc-masked window	notebooks/11 \$PowerBin	7 bins at target $S/N=15$; 2 outer bins still hit $\sigma > 400$ (irreducible KCWI S/N limit at aperture edge) — improvement vs Test 19 narrow-window failure but not fully cured	⚠ partial
K3	Wide vs narrow window for resolved maps	notebooks/11	Wide arc-masked unlocks per-spaxel + PowerBin maps; narrow window was per-spaxel-infeasible	✓
M. Reduction-pass systematic (2026-05-20)				
M1	New _mtwdo_ red-side reduction PROMOTED to headline 2026-05-26	notebooks/09e_new_ red_reduction.ipynb, scripts/run_wide_ sigma_e.py --cube new_raw_KCWI/New_red/ Nov17_2025_DESJ0206_ RL_combined_mtwdo_ icubes_wcs.fits	$\sigma_e(<R_e) = 265.76 - 18.80 / +18.33$ km/s (with new reduction-pass component). Per-SPS spread 2.67 km/s. Δ vs old reduction = +10.92 km/s → carried as ± 5.5 km/s reduction-pass systematic (M2). Cube co-aligned to $<0.002''$ with the old reduction (centering ruled out), shift consistent across all 15 polynomial degrees + all 3 SPS templates (mechanism is distributed across the wide window, not localized to one λ boundary — confirmed by 8000-Å boundary-mask cross-test M3). Caches: results/run_wide_sigma_e/new/. Figures: results/figures/nb09e_reduction_comparison.png, nb09e_grating_consistency_check.png. Audit: NOTES_nb09e_audit_2026-05-20.md (Addendum 2026-05-26).	✓ HEADLINE

#	Test	Where	Result	Status
M2	Reduction-pass systematic budget component	M1 vs the old-reduction headline	half- $\Delta = (265.76 - 254.85) / 2 = \pm 5.46$ km/s, carried as ± 5.5 km/s in the systematic budget (matches our peak-to-peak/2 convention from E5 / J10). Flag: only 2 reductions available; revisit / refine if a 3rd lands. Documented in METHODS_AND_SYSTEMATICS.md Part I.9 + project memory project_nb09e_reduction_systematic.md.	✓
M3	8000-Å boundary-mask cross-test to localize the +10.9 km/s shift	/tmp/test_8000A_boundary_mask.py, caches at results/run_wide_sigma_e/new_8000A_masked/	$\sigma_e = 268.73$ km/s after adding extra mask at def-rest 4715–4834 Å (= obs 7900–8100 Å) → hypothesis REJECTED (shift NOT localized to this one transition). Recovery toward headline = -27% . Confirms the +10.9 km/s shift is distributed across the wide window — likely from the 2-tier per-night flux scaling tilting the entire continuum, not from a single λ boundary.	✓ negative
M4	ppxf clean=True sigma-clip cross-test on new cube (N=100)	/tmp/test_sigma_clip_new_cube.py, caches at results/run_wide_sigma_e/new_sigma_clip_N100/	$\sigma_e = 265.80$ km/s (Δ vs headline = $+0.04$ km/s). 0 pixels rejected at every (sps, degree) of 45 fits (of ~2161 good pixels each). Reason: ppxf clean=True scales against the noise array; the noise array on this dataset is overestimated, so a 3σ threshold in noise units sits well above the actual outliers (see M5). Test by itself is uninformative; superseded by M5.	✓ negative

#	Test	Where	Result	Status
M5	Local-MAD bad-pixel cleaning (rolling 75-pix window, 3σ threshold) on new cube (N=100)	/tmp/test_local_mad_clip.py, caches at results/run_wide_sigma_e/new_local_mad_clip_N100/	52 outlier pixels flagged (of 2161 good); biggest is 6-pix cluster at def-rest 5232–5236 Å (obs 8768–8774 Å) at 26σ in local-MAD units — clear unresolved cosmic-ray residual that the noise-scaled clean=True missed. σ_e shifts to 267.18 km/s (Δ vs headline = +1.42 km/s), FSPS-dominated (+3.67). Carried as a stated sensitivity in METHODS Part I.9 (“What is not in the budget”); not folded into formal budget because the shift is well below stat 1σ (± 6.1).	✓ sensitivity
M6	OLD-cube replication of M5 clip (same 52 def-rest λ)	/tmp/test_local_mad_clip_OLD_cube.py, caches at results/run_wide_sigma_e/old_local_mad_clip_N100/	σ_e shifts 254.85 → 256.01 km/s (Δ = +1.16 km/s on OLD cube). Both cubes shift by $\sim +1.3$ km/s → 52 bad pixels are intrinsic to the data, not reduction-specific. Reduction-pass gap preserved (10.91 un-clipped → 11.17 clipped).	✓ replicates
M7	Noise-scaling + ppxf clean=True scan on new cube	/tmp/test_noise_scaled_clean.py, caches at results/run_wide_sigma_e/new_noise_scaled_clean_N100/	Scanned noise $\times\{1.0, 0.5, 0.3, 0.2, 0.1, 0.05\}$ at FSPS deg=22 → ppxf clips {0, 1, 18, 95, 591, 1247} pixels respectively. Full pipeline at noise $\times 0.3$ (19 pix/fit, closest to M5’s 52) gives $\sigma_e = 271.04$ km/s ($\Delta = +5.28$ km/s) — larger shift than local-MAD because ppxf clean=True is iterative and identifies a different (smaller) set of pixels. Sensitivity envelope: cleaning shifts range +1 to +5 km/s depending on method.	✓ sensitivity

#	Test	Where	Result	Status
M8	He I 3819 source-emission mask added to ARC_MASKS_REST (5237–5253 Å) (2026-05-27)	<code>scripts/run_wide_sigma_e.py --cube {new,headline}_clean_hei</code> , caches <code>results/run_wide_sigma_e/{new,headline}_clean_hei/</code>	Identified 3-pixel +ve residual cluster at def-rest 5244–5248 Å (= source-rest 3818–3820 Å, matches He I 3819.60 emission from $z=1.302$ source). Consistent across NEW + OLD reductions → astrophysical, not CR. Adding mask shifts NEW: 268.98 → 271.87 (+2.89), OLD: 260.44 → 264.16 (+3.72). Reduction-pass gap shrinks 8.54 → 7.71 → reduction-pass systematic refined $\pm 4.27 \rightarrow \pm 3.86$. New TOTAL sys = ± 17.25 ; new paper headline $\sigma_e(<R_e) = 271.87 \pm 17.86$ km/s (asym $-17.99/+17.74$).	✓ HEADLINE update
M9	def-rest 5193–5210 Å bump investigation — does NOT mask (it's the Mg b LOSVD wing)	<code>/tmp/smoke_extra_mask_5190_5210.py</code> (N=100 smoke); cube-arc spectrum extraction in <code>AGEL0206_arc_source_spectrum.pdf</code> ; sky-line noise diagnostic	Visible +5–7% bump in deflector data above ppxf fit at def-rest 5193–5204 Å. NOT source emission (arc-spaxel integrated spectrum in source-rest 3782–3788 Å is flat, no emission line — see fig). NOT sky residual (cube <code>noise_sky</code> shows 0.6–0.9× median std in this range — well below the strong OH airglow forest at obs 8760–8790 Å that's already covered by BAD_PIXELS + the He I mask). Diagnosis: Mg b LOSVD wing — the bump is mirrored on the blue side of Mg b (def-rest 5160–5170 also has +ve residuals). Mg b at 5183 Å convolved with $\sigma \approx 270$ km/s LOSVD broadens 4–5 Å → wings extend out to ~5190–5210 Å. Smoke test: masking 5190–5210 drops σ_e to 264.83 km/s (-7.04 km/s) — the LARGEST single mask effect seen. Verdict: DO NOT MASK — this region is the <i>signal</i> ppxf uses to constrain σ_e , not contamination. Documenting so we don't re-investigate.	✓ DO NOT MASK

#	Test	Where	Result	Status
M10	Full sky-line audit across fit window 3800–5400 Å (2026-05-27)	BAD_PIXELS_REST updated to 35 entries (26 original + 9 added). Caches <code>results/run_wide_sigma_e/{new,headline}_clean_hei/</code> ; per-band rationale inline in <code>scripts/run_window_sweep.py</code> . Identification via cube <code>noise_sky</code> thresholded at $>2.5\times$ median; each added band cross-checked against arc spectrum (AGEL0206_arc_source_spectrum.pdf) to confirm no source-emission counterpart.	Found 9 unmasked OH airglow / sky-residual bands in the fit window after user-flagged def-rest ~ 4600 Å structure: (4380.0-4384.0), (4553.0-4554.6), (4602.0-4610.0) $\leftarrow$ the 4600 Å feature, (4687.3-4688.3), (4691.5-4693.6) merged, (4770.8-4771.8), (4951.6-4955.0), (5011.9-5015.3), (5029.8-5033.8). All in def-rest Å, sky_noise $2.5\text{--}4\times$ median. NEW cube σ_e : 271.87 $\rightarrow$ 269.62 (-2.25 km/s). OLD cube σ_e : 264.16 $\rightarrow$ 262.72 (-1.44 km/s). Reduction-pass gap shrinks 7.71 $\rightarrow$ 6.90 $\rightarrow$ refined $\pm 3.86 \rightarrow \pm 3.45$. New TOTAL sys = ± 17.16 ; new paper headline $\sigma_e(<R_e) = 269.62 \pm 17.78$ km/s (asym $-17.92/+17.65$).	✓ HEADLINE update
M11	Rigorous re-derivation of carried I-shape, F200-mask, fit-window systematics on NEW cube + M10 masks (2026-05-27)	Caches <code>results/ishape_sweep_wR3800_5400_arcmask_M10/</code> (10 I-shapes $\times$ 3 SPS $\times$ N=250), <code>results/maskweight_sweep_wR3800_5400_arcmask_M10/</code> (3 weights $\times$ 3 SPS $\times$ N=500), <code>results/nb09a_wavelength_sweep_M10/</code> (3 windows $\times$ 3 SPS $\times$ N=500). Total compute ~ 1.5 h via <code>/tmp/run_full_systematics_barrage.sh</code> .	I-shape: range 266.83–271.37 km/s $\rightarrow$ peak-to-peak/2 = ± 2.27 (was ± 1.5 carried). F200 mask: w00=269.69, w50=261.95, w100=256.39 $\rightarrow$ peak-to-peak/2 = ± 6.65 (was ± 3.8 carried; bigger because cleaned NEW cube + Balmer-unmasked is more arc-sensitive). Fit-window: wR3800_5400_arcmask=269.66, wR4000_5400_arcmask=268.58, w6500_7500=276.22 $\rightarrow$ peak-to-peak/2 = ± 3.82 (was ± 15 carried; cleaned NEW cube + M10 brings wide/narrow into agreement). Net: sys total $\pm 17.16 \rightarrow \pm 10.81$; TOTAL $\pm 17.78 \rightarrow \pm 11.77$ (drop of -6.0 km/s). NEW HEADLINE: $\sigma_e(<R_e) = 269.62 \pm 11.77$ km/s (asym $-11.98/+11.57$).	✓ HEADLINE update

#	Test	Where	Result	Status
M12	R_e-source fold-in + masking-approach + H δ cross-checks (2026-06-08, N=500)	scripts/run_sigma_e_ Re_systematic_wide. py (nb15), run_sigma_e_mask_ systematic.py (nb13), run_sigma_e_hdelta_ test.py (nb16); registry scripts/paper_ values.py→results/ PAPER_VALUES.json	(1) D7 R_e-source folded in: ± 6.13 (wide-window, 4 estimators; user chose full spread 2026-06-08) → sys $\pm 10.81 \rightarrow \pm 12.43$; HEADLINE $\sigma_e = 269.62$ ± 13.27 km/s (asym $-13.45/+13.10$). (2) Arc-masking-approach systematic = ± 5.85 (ex- pert/serisic/perband/global reprojected to IFU grid; spectroscopic analogue of the M $\star$ ± 0.16 dex term) — overlaps F200-mask ± 6.65 , NOT added (larger-of-two). (3) H δ : KEEP UNMASKED — local-MAD shows H δ well-fit (max resid/noise $0.44 < \text{global } 0.81$), masking shifts $\sigma_e +6-8$ km/s = LOSVD information not contamination (M9 pattern); TODO in bootstrap_ppxf.py closed.	✓ HEADLINE update
L. Figure preparation (2026-05-13)				
L1	Figure 2 (narrow) — single posterior inset	AGEL_0206_ApJL_ Figures/figures. ipynb cell a546db7f, Mbh_sigma_SED_final. pdf (was AGEL0206_sigma_e_SED_final.pdf)	Title shows $\sigma_e = 267 \pm$ $24(\text{stat}) \pm 18(\text{sys})$ km/s; no-arc-mask overlay removed	✓
L2	Figure 2 (wide, headline) — single posterior inset	figures.ipynb cell fig2_wide, AGEL0206_sigma_e_ SED_final_wide.pdf	$\sigma_e = 255 \pm 6(\text{stat}) \pm$ $17(\text{sys})$; residuals panel; 9 absorption features labeled; 4 arc-emission masks shaded	✓
L3	Figure 3 (M BH -M star) cleanup	figures.ipynb cell fig3_clean_no_err, Mbh_Mstar_relation_ clean.pdf	No per-object error bars; uniform color for local sample; filtered to Greene+2020 E+S0 list (60 unique, 4 KH13 suffix variants matched; 19 post-KH13 extras dropped)	✓
L4	Figures 2 + 4 updated to principled-mask M $\star$ (2026-05-29)	figures.ipynb cells 19/35/36; AGEL0206_sigma_e_ SED_final_wide.pdf, Mbh_Mstar_ relation{,_clean}. pdf	Fig2 inset posterior → perband_raw_10pct (median 11.16) + asymmetric title 11.16 $+0.32/-0.08$ + shaded fill-in reach to 11.46; Fig4 AGEL point → $10^{11.16}$, asymmetric xerr $-0.08/+0.32$. Backup figures.ipynb.bak. Mstar_principled_ 2026-05-29	✓

N — Photometry arc-mask verification + principled masking (2026-05-29) |||||

N1 | F200LP hand mask reproducible from objective selectors | scripts/arc_mask_verification.py,

notebooks/12 Part I | Color (m_F200LP–m_F140W) + Sérsic-residual reproduce the expert F200LP mask; Sérsic-residual (k=3) matches expert photometry to 0.016 mag, R_e to 3% | ✓ |

N2 | S/N-regime sweep (over-masking quantified) | scripts/arc_mask_verification.py, fig arcmask_snr_sweep.png | snr∈{2..20}, k∈{2..8}; F200LP model-flux-fraction stays 0.03–0.10 (clean), Δ mag saturates at $k \approx 3$ = natural stopping point | ✓ |

N3 | Independent per-band Sérsic OVER-MASKS F140W/JWST | scripts/principled_mask_photometry.py, figs principled_*.png | Single Sérsic under-fits bright IR galaxy (median resid +1.0) → over-masks 0.21–0.36 mag; low mdlfrac (0.02–0.03) is a FALSE guard. Use F200LP-reproj instead | ✓ negative |

N4 | Reprojected F200LP footprint = transferable standard | scripts/mask_method_comparison.py, figs maskcompare_*.png | Reproducing expert JWST mags to 0.01–0.02 mag by reprojecting F200LP (NOT union with F140W — F140W mask covers core, 17px at $r < 0.4''$) | ✓ |

N5 | 2-component (bulge+disk) Sérsic model | scripts/photometry_systematics.py:fit_2comp | Single Sérsic under-fits JWST; 2-comp cuts galaxy-body RMS 3.3–3.5× (F150W2 3.25×, F322W2 3.46×). No PSF (env lacks webbpsf) — flagged | ✓ |

N6 | IR source extension + per-band vs global mask | scripts/photometry_systematics.py, figs photsys_*.png | Deeper IR extends arc (F150W2 +9119px). raw is mask-size-dependent (up to 0.78 mag per-band↔global); filled is mask-def-independent (≤ 0.06 mag). Fill-in correction +0.18–0.96 mag | ✓ |

N7 | M★ budget — raw-central, one-sided-up systematic | results/photometry_systematics_Mstar.npz, fig Mstar_budget.png, notebooks/12 Part II | $\log M_\star = 11.16 \pm 0.08(\text{stat}) + 0.31(\text{sys})$ at 10% [$11.04 \pm 0.14 + 0.32$ at 20%]; sys = Sérsic fill-in (under-arc light), reach 11.46/11.36. per-band vs global negligible for filled. NEW HEADLINE M★ (supersedes 11.33) | ✓ HEADLINE update |

N8 | Explicit masking-approach systematic on M★ | results/Mstar_masking_systematic.npz, fig Mstar_masking_systematic.png | ± 0.16 dex (10%: ± 0.160 , 20%: ± 0.162) = peak-to-peak/2 across {expert-aper, per-band/global × raw/filled} spanning 11.15–11.47. Decomp: under-arc(raw↔filled) ± 0.15 , mask-def(per-band↔global) ± 0.004 , mask-extent(expert↔IR-ext) 0.18. Symmetric statement of the headline’s one-sided +0.31. TODO: σ_e analogue | ✓ |

N9 | PSF-convolved fill model (firm up the fill leg) | scripts/psf_fill_model.py, results/psf_fill_model.npz | Gaussian PSF at instrument FWHM (F200LP 0.08"/F140W 0.14"/F150W2 0.06"/F322W2 0.13"; env lacks webbpsf). Δ PSF(filled) ≤ 0.004 mag all bands → $\Delta M_\star \approx 0.01$ dex, negligible vs stat ± 0.08 / masking ± 0.16 . Fill is PSF-robust (arc outside PSF core) | ✓ |

A3b | R_e pixscale fix in measure_Re.hst_Re (diagnostic) | scripts/measure_Re.py (now reads WCS; F200LP literal 0.08→0.05) | F200LP cutout is 0.05"/pix (was hard-coded 0.08) → diagnostic F200LP R_e 3.05→1.91"; F140W unchanged. Headline $R_e=2.305''$ UNAFFECTED (final_sigma_e.py already WCS-correct: F200LP=2.52", F140W=2.16", mean $\approx$ 2.34"). Flag: two CoG algos differ (measure_Re 1.91 vs final_sigma_e 2.52 for F200LP) — pre-existing | ✓ fix |

A3c | CoG-algorithm reconciliation (measure_Re vs final_sigma_e) | notebooks/14_CoG_reconciliation.ipynb, scripts/measure_Re.py (hst_Re now r_max_arcsec=6.0 default), results/Re_cog_reconciliation.npz | The two CoG algos are the same up to r_max: measure_Re capped at 4.0" (arange(1,80,1) px), final_sigma_e integrates to 6.0". Matched r_max → agree $< 0.04''$ (residual = centroid_2dg vs fixed-RA/Dec center). Headline $R_e=2.305''$ = canonical (final_sigma_e, r_max=6"). New flag: raw CoG R_e is r_max-non-convergent (non-zero sky pedestal); headline sits at the top of a 2.1–2.5" method family (sky-sub 2.14", Sérsic 2.18") → R_e method systematic $\pm 0.08''$. Not folded (would shift σ_e aperture + M★; flagged for user). | ✓ reconciled |

1. Headline numbers

σ_e at three apertures (paper headline = wide arc-masked window)

Aperture	σ_e [km/s] (WIDE, headline)	σ_e [km/s] (NARROW, cross-check)	Notes
$R < R_e (=2.305'')$	255 ± 18 (254.85 ± 17.87)	268 ± 30 (267.95 ± 30.10)	paper headline — KH13/Greene+20 σ_e at WIDE; NARROW cross-check at 0.4σ
$R < R_e/2 (=1.152'')$	~ 225 (nb07c §6cum NARROW)	224 ± 18	gradient diagnostic (still cited from nb07c)
$R < R_e/8 (=0.288'')$	dropped	dropped	inside seeing $\text{FWHM}/2 = 0.64''$

Error-budget composition ($R < R_e$) — both windows side by side

Component	WIDE wR3800_5400_arcmask	NARROW w6500_7500	Source
Statistical (bootstrap pooled 1σ)	± 6.1	± 23.9	wild-bootstrap N=500, FSPS+EMILES+XSL pool
I-shape spread (10 shapes, post-Sersic2D-bound-fix)	± 1.5	± 3.7	results/ishape_sweep_wR3800_5400_arcmask/ + results/annular_bootstrap_07c_ishape/
F200 mask on/off Δ (peak-to-peak / 2)	± 3.8	± 7.5	results/maskweight_sweep_wR3800_5400_arcmask/ + results/mask_weight_sweep.npz
Frame fix (max σ shift across SPS, carried)	± 5	± 5	NOTES addendum + audit 1
Centering (5-center sweep $\pm 0.4''$, carried)	± 4	± 4	NOTES_centering_investigation_2026-04-27.md
Fit-window (3-window spread from §9)	± 15	± 15	nb09 §9: w6500_7500 / wR3800_5400_arcmask / wR4000_5400_arcmask (orthogonal $H\beta$ +Mg b)
Quadrature total	± 17.87	± 30.10	results/sigma_e_final_systematics_nb09d.npz

Where each component changed between windows

- Statistical $\pm 24 \rightarrow \pm 6.1$ ($4\times$ tighter): wide window has $\sim 4\times$ more spectral pixels (2161 vs 555 good pixels). This is the headline reason to elevate the wide window as primary.
- I-shape $\pm 13 \rightarrow \pm 1.5$ ($9\times$ tighter): driven mostly by the Sersic2D bound-fix (J7), which removed the $n=0.30$ pathology that contributed ± 5.4 of the old narrow ± 13 . The wider window also stabilizes σ across I-shape choices.
- Mask $\pm 16 \rightarrow \pm 3.8$ ($4\times$ tighter): the spectral arc-emission masks (J3) absorb most of what the F200 spatial mask used to absorb; the F200 mask becomes a sub-budget perturbation rather than a primary uncertainty.
- SPS template spread: unstated as a separate line in the budget because it's contained in the statistical pool. But it collapsed $26 \rightarrow 4$ km/s between windows (J5) — was a major budget item at the narrow window, sub-statistical at the wide.
- Frame fix, centering: unchanged (carried from the narrow-window era's careful audits — physically independent of the fit window).
- Fit-window ± 15 : NEW budget component (didn't exist before nb09d because there was only one window). Spread between three physically meaningful windows: Ca H+K + G (narrow), full canonical range (wide arc-masked), and $H\beta$ + Mg b only (wide minus Ca H+K). This is now the dominant residual systematic.

Sensitivity values (for paper text)

WIDE arc-masked window (headline): - No-spatial-mask ($w=1.0$) at $R < R_e$: $\sigma_e = 247.28$ km/s — Δ from headline ($w=0$) = -7.6 km/s - Half-spatial-mask ($w=0.5$) at $R < R_e$: 250.15 km/s — Δ = -4.7 km/s - Per-SPS at wR3800_5400_arcmask: FSPS=253.9, EMILES=253.3, XSL=257.5 $\rightarrow$ spread 4.2 km/s

NARROW Ca H+K + G window (cross-check, retained from nb07c era): - Soft-mask ($w=0.5$) at $R < R_e$: 258.5 km/s — Δ from narrow-headline = -9.3 km/s - No-mask ($w=1.0$) at $R < R_e$: 252.8 km/s — $\Delta = -15.0$ km/s - $\sigma_e(w)$ is essentially linear with weak super-linearity (quadratic $c=+7.3$) - Per-SPS at $w6500_7500$: FSPS=253.7, EMILES=267.9, XSL=279.6 → spread 26.0 km/s (the SPS systematic that motivated the move to wide)

2. Detailed test catalog

This expands each row of the matrix above. Subsections reference scripts, notebooks, and result files — open the linked file for the detailed code and the inline plots/output.

A. Foundational data and geometry

A1 KCWI cube provenance — The cube `Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits` is the final multi-night reduction stacked by K. R. Gupta on UT 2026-01-07. The header's `DATE-OBS = 2025-11-17`, `PROGNAME = U002`, `PROGPI = Jones` describes only the first input frame (`kr251117_00129`), not the full input set.

Verified contributing nights (cross-checked 2026-05-26 against the Keck observing logs):

Night (UT)	Program	PI	DESJ0206 frames	On-source RED + BLUE
2025 Aug 30	K409	TBD	12 RED kr250830_00090-00101, 4 BLUE kb250830_00052-00055	60 min + 66 min
2025 Nov 17	U002	Jones	20 RED kr251117_00129-00148, 5 BLUE kb251117_00087-00091	90 min + 110 min
2024 Dec 29	TBD	(Yuguang Chen)	4 RED kr241229_00092-00095 per local stack list	NOT confirmed as DESJ0206 pointings — only the 44 MB image-only PDF kcwi-241228-written- log.pdf (Drive id 16zW08yaCIvRtoCGenNTndvB7dYdrCkMH) exists; no machine-readable log. Pending raw-frame header dump.

Confirmed total on-source (Aug 30 + Nov 17): ≈ 170 min RED + 176 min BLUE.

Two earlier-recorded contributors are incorrect: ~~Sept-29-2025 K409~~ has zero DESJ0206 entries in its log; ~~Dec-29-2024~~ remains unverified. Cite K409 + U002 in the paper, flag Dec 29 as pending. See `reference_kcwi_data_properties.md` (memory) for full Drive IDs and log-file pointers.

A2 Systemic redshift — Independent line-by-line Gaussian fitting of Ca H+K, [OII], G-band, etc. gives $z_{\text{systemic}} = 0.67564$ (NIST air wavelengths). DR2 catalog $z=0.67511$ (used by older notebooks 03/03b) is offset by ~ 95 km/s from the line-fit value; nb09 uses 0.67564 throughout. Notebook: `notebooks/04_redshift_verification.ipynb` Module: `scripts/redshift_verify.py` (provides `ABSORPTION_LINES` / `EMISSION_LINES` dicts, `gaussian_fit`, `multi_line_fit`, `oii_doublet_fit`)

****A3 Effective radius $R_e = 2.305''$ — Mean of F140W and F200LP masked curve-of-growth values (2.168 and 2.441'').** The masked CoG zeros mask-flagged HST pixels before the radial integral so the arc/companion contributions don't bias the bright-end half-light point. Computed in `scripts/measure_Re.py` and re-derived in `scripts/final_sigma_e.py:curve_of_growth` for the nb09 pipeline. Cross-check: IFU white-light proper-masked CoG gave 2.61'' earlier — the HST-based 2.305'' is now the headline.

A4 HST-mean center — `photutils.centroid_2dg` on F140W and F200LP cutouts (with their own `_mask.fits` as the mask) gives sub-pixel centroids in each HST WCS. We average in world coordinates, then propagate to IFU sub-pixel: `scripts/final_sigma_e.py:find_center`. F140W vs F200LP centroid offset = 0.36" — flags the F200LP arc-affected detection but the HST-mean is robust (verified by the 5-center sweep, F1).

A5 Stellar mass — Bagpipes SED fitting on HST + JWST aperture photometry (notebook 02, $N_{\text{posterior}} = 500$). Bagpipes config: exponential- τ SFH, BPASS+CLOUDY templates, Calzetti dust, free metallicity, redshift fixed at 0.67564.

Per-filter observed flux + best-fit modelled flux:

Filter	$\lambda_{\text{pivot}} [\text{\AA}]$	AB obs	AB model (p50)	F_obs [10^{-18} cgs]	F_model (p50)	residual
F200LP	4 972	22.613	22.563	3.97 ± 0.06	4.16	+4.7%
F140W	13 923	19.134	19.264	12.47 ± 0.05	11.06	−11.3%
F150W2	16 720	18.942	19.033	10.31 ± 0.005	9.49	−8.0%
F322W2	32 470	18.604	18.505	3.73 ± 0.001	4.09	+9.6%

Posteriors:

Parameter	p16	p50	p84
$\log M_{\star}/M_{\odot}$	11.24	11.33	11.41
mass-weighted age [Gyr]	3.69	5.10	6.11
exponential SFH age [Gyr]	4.46	5.98	6.97
exponential τ [Gyr]	0.43	0.68	1.21
metallicity $Z/Z_{\odot}$	0.17	0.89	1.87
A_V [mag]	0.34	0.82	1.55
z_{phot}	0.674	0.675	0.676

→ $M_{\star} = 2.15 \times 10^{11} M_{\odot}$; z_{phot} agrees with the line-fit $z = 0.67564$ within ± 0.001 .

The model under-predicts the two NIR bands (F140W, F150W2) by ~8-11% and over-predicts the bluest+reddest by ~5-10%. Classic “redder NIR slope than a single- τ SFH prefers” tension; not catastrophic, but flags that a more flexible SFH (delayed- τ or non-parametric) might tighten the fit if revisited.

Sersic-total cross-check (notebooks/08_sersic_total_photometry.ipynb, `scripts/measure_Re.py`-style 2D Sersic fits per filter, extrapolated to total flux):

Filter	AB aperture	AB Sersic-total	Δmag	Sersic n	$r_{\text{eff}} ["]$
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Re-fed through Bagpipes with the same priors:

Quantity	aperture (paper)	Sersic-total	Δ
$\log M_{\star}/M_{\odot}$ p50	11.33 +0.07/−0.09	11.40 +0.11/−0.15	+0.065 dex
$M_{\star} [M_{\odot}]$	2.15×10^{11}	2.49×10^{11}	+16%

The aperture under-counts outer-profile flux by ~0.8-1.9 mag depending on filter, but at the integrated-mass level the Sersic correction is +0.07 dex — within the aperture-photometry quoted uncertainties. The paper cites the aperture value as the headline; Sersic-total is the sensitivity floor.

Cache: results/bagpipes_sed_results.npz (aperture, 500-sample posterior), results/bagpipes_sersic_refit.npz (Sersic-total), results/sersic_total_photometry.npz (Sersic fit parameters). Figure: results/AGEL0206_spectra_SED_fit.pdf.

B. Pipeline correctness audits (scripts/audit_ppxf_methodology.py)

Four orthogonal correctness audits run end-to-end on the headline aperture spectrum to verify ppxf usage matches Cappellari (2017, 2023). Saved in results/ppxf_methodology_audit.npz. Documented in NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md ADDENDUM 2026-04-29.

B1 Instrumental LSF — KCWI header DISPSCAL = 0.294 (instrumental σ in pixels) $\times$ CD3_3 = 1.000 Å/pix $\rightarrow$ FWHM_inst = $2.355 \times 0.294 = 0.692$ Å (constant in observed Å). At the rest-frame fit window (3879–4476 Å), σ_v _inst $\approx$ 12–14 km/s. Galaxy $\sigma \approx$ 270 km/s is resolved by $\sim 20\times$. Diagnosed in scripts/ifu_spectral_resolution.py.

B2 σ _inst sensitivity — Sweep DISPSCAL $\times$ {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Max $|\Delta\sigma| = 0.83$ km/s across all SPS. Reason: ppxf's sps_lib clips fwhm_diff² = (FWHM_gal² – FWHM_tem²).clip(0) (sps_util.py:169). When templates are intrinsically broader (FSPS, EMILES at our band), no convolution is applied — σ becomes insensitive to DISPSCAL. LSF is rock-solid. Script: scripts/sigma_inst_sensitivity.py

B3 SPS template frames — Located the Ca K minimum in each shipped template: - FSPS spectra_fsps_9.0.npz: Ca K at 3934.86 Å $\rightarrow$ +1.20 Å vs air rest = vacuum - EMILES spectra_emiles_9.0.npz: 3933.82 Å $\rightarrow$ +0.16 Å = air - XSL spectra_xsl_9.0.npz: V_sys closes at +9 km/s with air-conversion $\rightarrow$ air End-to-end V_sys closure (NOTES Test 3) confirms.

B4 V_sys air vs vacuum $\times$ 5 polynomial degrees — Run ppxf at degrees {15, 18, 21, 24, 27} for each SPS in both frames. $\Delta V(\text{air} - \text{vac}) \approx +83$ km/s for FSPS, ≈ -82 km/s for EMILES & XSL, consistent within ± 2 km/s across degrees. Frame identification is robust.

B5 $z \times$ air-vac differential — Apply vac_to_air at observed wavelengths (KCWI 6500–7500 Å) $\rightarrow$ v_offset = -82.7 km/s. Apply at rest wavelengths (3879–4476 Å) $\rightarrow$ -84.5 km/s. Differential = 1.83 km/s — sub-budget. Either convention is defensible; we follow Cappellari's pattern (apply at obs).

B6 fwhm_gal_dict frame check — Confirmed _prep_spectrum_for_ppxf builds the dict as {"lam": lam_gal_rest, "fwhm": fwhm_gal_rest} in REST frame, matching ppxf_example_high_redshift.py:99–101. FWHM_obs = 0.692 Å $\rightarrow$ FWHM_rest = 0.413 Å at $z=0.67564$. sps_util.py:167 interpolates this onto template lam_temp (also rest), then varsmooth pre-broadens.

C. SPS templates and frame-fix

C1 Frame-aware ppxf — scripts/bootstrap_ppxf.py exposes frame_galaxy='auto' which reads SPS_NATIVE_FRAME = {fsps: vacuum, emiles: air, xsl: air}. The galaxy spectrum is prepared in the matching frame (util.vac_to_air()) applied as scalar median when needed, matching ppxf_example_kinematics_sdss.py:127–134). Diagnosed Apr 2026 — pre-fix the pipeline always converted to air, producing a -90 km/s FSPS V_sys offset.

C2 V_sys closure — Same galaxy spectrum, swap only the frame. Pre-fix: FSPS -90 , EMILES +3, XSL +9. Post-fix (frame-aware): FSPS -7 , EMILES +3, XSL +9. Split-track collapsed from $\sim 110 \rightarrow \sim 15$ km/s. NOTES Test 3.

C3 σ shift from frame fix — Side-effect on σ : ≤ 5 km/s per SPS, ≤ 2 km/s on the 3-SPS pool. Carried as ± 5 km/s in the error budget (ironically dominated by the per-SPS shifts that pool out, but conservative).

C4 3-SPS pooled posterior — scripts/final_sigma_e.py:pool_sps concatenates per-SPS bootstrap σ samples after subtracting per-SPS V_sys medians (so V offsets don't inflate the V posterior; σ posteriors are concatenated raw — no V_sys correction needed for σ). The headline $\sigma_e(<R_e)$ is the 16/50/84 percentile of this combined-SPS pool.

D. Aperture and I(r) framework

D1 §6cum cumulative I-weighted ppxf — For each spaxel inside $R < R_{\text{max}}$, build I-weight from the IFU 6500–7500 Å white-light band, drop arc-mask-flagged spaxels, normalize and sum to a single 1-D spectrum, fit ppxf. This is the headline path — matches KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA σ_e definition (Capellari+2006 eq. 1). Cross-references in `7.claude/.../reference_cumulative_vs_annular_sigma_e.md`. Code: `scripts/final_sigma_e.py:run_aperture_sps`.

D2 §6cum vs §7 (annular) — §7 discrete Gültekin annular sum at $R < R_{\text{safe}}$ gives 254.99 km/s; §7b flat- σ extrapolation gives 271 km/s; both within 1σ of §6cum’s 267.82 km/s. §6cum is the headline because (a) it matches the literature definition, (b) it preserves LOSVD line-shape information that §7’s moment-pooling discards, (c) bright-center I-weighting auto-suppresses arc contamination.

D3 I(r) shape sweep (10 shapes) — Hold the spaxel selection fixed (F200-raw mask, $R < R_e$), vary only the I-weight map. Shapes: 1. IFU_band — 6500-7500 Å (headline) 2. IFU_wl — full IFU white-light 3. F140W_raw — HST F140W reprojected 4. F200LP_raw — HST F200LP reprojected 5. F140W_arcmasked — F140W with arc zeroed 6. F200LP_arcmasked — F200LP with own arc-mask zeroed 7. F140W_1Dcog — annular-mean F140W → interp1d to spaxels 8. F200LP_1Dcog — annular-mean F200LP → interp1d 9. F140W_Sersic2D — full 2D Sérsic fit 10. F200LP_Sersic2D — full 2D Sérsic fit (poor fit, $n=0.30$ — outlier)

Range excluding F200LP_Sersic2D outlier = 12.9 km/s → ± 13 km/s budget. Script: `scripts/run_isource_shape_sweep.py` Analysis: `scripts/analyze_isource_shape_sweep.py` Caches: `results/annular_bootstrap_07c_ishape/{shape}_{sps}.npz` Figure: `results/figures/nb07c_ishape_sweep.png`

D4 I-shape sweep refresh post-frame-fix — Original sweep at $N=250$ predated the frame fix. Re-ran at $N=500$ with `frame_galaxy='auto'` on 2026-04-29. Per-SPS σ shifts < 0.5 km/s; spread unchanged at 12.9 km/s; ± 13 budget validated. Pre-frame caches preserved at `results/annular_bootstrap_07c_ishape_oldframe/`. NOTES addendum (d).

D5 Aperture choice — Three apertures considered: $R < R_e/8$ ($=0.288''$), $R < R_e/2$ ($=1.152''$), $R < R_e$ ($=2.305''$). $R < R_e/8$ dropped because it sits inside half the seeing FWHM ($=0.64''$) — only 3 spaxels, and the inner σ would underestimate due to seeing-broadened sampling. $R < R_e$ is the paper headline; $R < R_e/2$ is quoted as a gradient diagnostic.

D6 Equal-N vs equal-width annular binning — For the §7 cross-check only. Equal-N inside $R_{\text{safe}} = 3R_e/4 + 1$ outer flagged bin (`notebooks/07c`) gives balanced bootstrap variance per bin and isolates arc contamination in the outer bin. Equal-width annuli (`notebooks/07b`, 5 bins) shifted σ_e by ~ 10 km/s — at the SPS-systematic level (± 24). Equal-N is the §7 paper choice.

D7 R_e source systematic (mean vs F140W vs F200LP vs Ca H+K + G) — The headline integrates the I-weighted aperture spectrum inside the mean F140W+F200LP masked CoG R_e ($=2.305''$). Three alternative R_e definitions test sensitivity to that choice (Track A masked, $N=500$):

R_e source	R_e ["]	σ_e [km/s]	$\Delta\sigma_e$
mean (paper)	2.305	267.82	—
F140W only	2.168	264.87	−2.96
F200LP only	2.441	275.86	+8.04
Ca H+K + G-band depth	2.866	281.74	+13.92

Spread = 16.9 km/s — at the mask-budget level (± 16) but sub-budget vs the total ± 32 . σ_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). The Ca H+K + G-band absorption- depth I-map (rest 3925-3942, 3960-3976, 4297-4313 Å, summed (continuum − flux) per spaxel — see `cahk_g_line_depth_map` in `scripts/final_sigma_e.py`) is the most stellar-only of the four; its larger R_e reflects that the deflector light extends further than the F200LP UV-leaning band suggests. Code: `scripts/final_sigma_e.py §7 + cahk_g_line_depth_map`. Display: `notebooks/09 §7b + results/figures/nb09_re_source_systematic.png`. Caches: `results/final_sigma_e_paper/Re_{F140W,F200LP,CaHK}_{sps}_N500.npz`.

E. Mask treatment

Masking vs no-masking — pro/con summary

Track	Pro	Con
Hard mask ($w=0.0$, headline)	Removes arc contamination; matches literature convention; F200LP mask is purpose-built for this lens	Drops 38 of 184 spaxels inside $R < R_e$ (~21% by count, ~27% by I-weight) → lower S/N; hard boundary may over-clip a few edge spaxels
No mask ($w=1.0$, sensitivity)	Maximum S/N; no boundary artifacts	Arc adds low-velocity continuum dilution → σ biased low by 15-17 km/s ($\Delta_{\text{mask}} = -16$ km/s at $R < R_e$)
Soft mask ($w=0.5$)	Smooth middle ground	Super-linear in w (quadratic $c=+7.3$); threshold-dominated bias → first 25% of arc weight already captures 34% of total mask sensitivity. Arbitrary weight choice; not a literature convention
Mask dilation (E6)	—	Inflates σ via noise (dropped spaxels are S/N contributors, not bias contributors). Negative finding — do not dilate

The 16 km/s no-mask Δ is propagated into the error budget as $\sigma_{\text{mask}} = \pm 16$, so the choice is bounded, not assumed.

E1 F200LP arc mask reprojection — F200LP_mask.fits (HST 0.04"/pix, arc-tuned) is reprojected to the IFU 0.30"/spax grid via `scipy.ndimage.map_coordinates(order=0)` (nearest-neighbor). 69 of 10000 spaxels (0.69%) flagged. Inside $R < R_e$: 38 of 184 spaxels in the aperture, ~21% — but only ~27% of the I-weight (arc spaxels are mostly off-center).

E2 Hard-mask ($w=0.0$) headline — track A. E3 No-mask ($w=1.0$) sensitivity — track B. $\Delta_{\text{mask}} = -15.0$ km/s. E4 Soft-mask single point ($w=0.5$) — track C. $\Delta_{\text{soft}} = -9.3$ km/s. E5 5-point mask-weight sweep ($w \in \{0, 0.25, 0.5, 0.75, 1\}$) — Per-step σ drops at $R < R_e$: $5.0 \rightarrow 4.3 \rightarrow 3.5 \rightarrow 2.2$ km/s. Quadratic $c = +7.3$ (concave-up, super-linear). Threshold-dominated bias: a few heavily-arc-contaminated spaxels carry most of the bias. First 25% of arc weight captures 34% of the total mask sensitivity. Scripts: `scripts/soft_mask_track.py`, `scripts/analyze_mask_weight_sweep.py` Figure: `results/figures/nb09_mask_weight_sweep.png` NOTES addendums (b) and (c).

E6 Mask dilation — Tested whether expanding the F200 mask further reduces σ . Negative finding: dilation inflates σ via noise — the dropped spaxels are S/N contributors, not bias contributors. Memory: `7.claude/.../project_nb07d_overmasking_finding.md`. Notebook: nb07d.

E7 Arc-spectrum subtraction sibling — Build an arc-only spectrum from F200-masked spaxels, fit $\alpha \times \text{arc} + \beta \times \text{deflector}$ model, subtract. Result matches §6cum within 0.1 km/s → *residual* arc dilution (the small fraction that leaks through the F200 mask) is sub-dominant at production statistics. The F200 mask already does the heavy lifting. Note: this test compares “masked + arc-subtract” against “masked alone” — it does NOT directly validate the no-mask vs masked Δ mechanism (that’s E8). Notebook: nb07e.

E8 Arc-as-sky mechanism test (rigorous test of the no-mask Δ) — Run §6cum at the *no-mask* $R < R_e$ aperture with the bright outer-arc spectrum passed to ppxf as a sky template (free-amplitude additive component, NOT convolved with the deflector LOSVD — physically appropriate since the arc is at $z=1.302$, decoupled from the deflector kinematics). At $N=500$, FSPS+EMILES+XSL pooled:

Track	σ_e [km/s]	Δ from σ_A
A: masked headline	267.82	—
B: no-mask	252.82	-15.00
D: no-mask + arc-sky (E8)	274.64	+6.82

Recovery fraction $(\sigma_D - \sigma_B)/(\sigma_A - \sigma_B) = 145\%$ — i.e. arc-sky overshoots the masked headline by ~ 7 km/s. Interpretation:

- Continuum dilution explains the FULL no-mask Δ . There is no detectable kinematic-blend or template-mix mechanism contributing in the *opposite* direction.
- The 7 km/s overshoot is consistent with the known caveat that the outer-arc-mask spaxels ($R > 3R_e/4$) still receive seeing-blurred deflector light at the few-percent level, so subtracting $\alpha \times \text{arc}$ oversubtracts a small amount of *real* deflector flux too. This inflates σ slightly.
- The masked headline (Track A, paper number) remains the cleaner measurement because it doesn't have this oversubtraction artefact.
- $\alpha_{\text{arc}} \approx 0.31$ across all three SPS — internally consistent.

Code: `scripts/run_07f_arc_sky.py` (uses `setup_ppxf_inputs_from_spectrum + ppxf sky=` argument). Display: `notebooks/07f_arc_sky_subtract.ipynb`. Caches: `results/arc_sky_07f/{sps}_N500.npz + results/arc_sky_07f.npz`. Figures: `results/figures/nb07f_{arc_template,recovery,posteriors}.png`.

F. Centering

F1 5-center sweep — Tested HST_mean (paper choice), F140W only, F200LP only, IFU peak, arc-suppressed IFU peak. σ_e spread = 3.7 km/s. Carried as ± 4 km/s in the error budget. Notes: `NOTES_centering_investigation_2026-04-27.md`.

F2 F140W vs F200LP centroid — $\Delta = 0.36''$ — driven by the F200LP arc dragging that filter's centroid toward the arc. F140W is the cleaner bulge centroid; HST-mean is the robust compromise.

G. Bootstrap and polynomial degree

G1 Wild bootstrap — Hybrid Rademacher sign-flip $\times$ local-residual scaling (75-pix rolling window). For each iteration: residual = (galaxy – bestfit) $\times$ sign, scaled by local σ . Re-fit, record V/σ . Implemented in `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`.

G2 Parallel bootstrap — `joblib + threadpoolctl` pinning BLAS=1 per worker (otherwise BLAS thread oversubscription pathology — saw 4-hour runtimes). 8 workers gives ~ 6 min for 6 fits at $N=500$. Script: `scripts/bootstrap_ppxf_parallel.py:run_bootstrap_single_degree_parallel`.

G3 Polynomial degree sweep — Degrees 15-29 (15 values) per fit. Diagnostic: σ vs degree should be flat within bootstrap envelope; monotonic trend indicates the polynomial is absorbing LOSVD signal. Plotted at `nb09 §6.5` and saved to `results/figures/nb09_sigma_vs_degree.png`. Saturation confirmed at this degree range — picked from earlier `notebooks/03c` analysis.

G4 $N=50$ vs $N=500$ — $N=50$ smoke and $N=500$ production agree on σ_e to within 1 km/s. $N=50$ is ~ 6 min, $N=500$ is ~ 50 min. Production headline is $N=500$; the smoke is for quick rebuilds.

H. Cross-checks

H1 §7 discrete Gültekin (refreshed 2026-05-01) — `notebooks/07c §7`, recomputed with frame-aware ppxf via `scripts/refresh_07c_gultekin.py`. Annular sum $\sigma_e^2(<R) = \Sigma F_j (V_j^2 + \sigma_j^2) / \Sigma F_j$ with arc-filter on outer annulus.

	Pre-frame-fix (Apr 24)	Post-frame-fix (May 1)	Δ
σ_e (arc-filtered)	254.99 $-24.2/+28.4$	256.17 $-13.0/+12.7$	+1.18

$\Delta < \pm 5$ km/s frame budget. Errors halved because the V_{sys} split collapsed from ~ 99 km/s (pre-fix) to ~ 8 km/s (post-fix per-SPS V_{sys} : `fsps` -11.2 , `emiles` -7.4 , `xsl` -3.1), shrinking the V^2 contribution to the Gültekin sum's MC scatter.

H2 §7b flat- σ extrapolation (refreshed 2026-05-01) — `notebooks/07c §7b`, same refresh. Push the §7 sum into the outer flagged bin assuming $\sigma_{\text{outer}} = \sigma$ at the last clean annulus.

	Pre-frame-fix	Post-frame-fix	Δ
σ_e	271 $-33/+35$	274.37 $-16.2/+17.4$	+3.37

Both H1 and H2 remain consistent with the §6cum headline (267.82) at $<1\sigma$. The point of these cross-checks is preserved by the refresh, and the post-fix bands are tighter, strengthening rather than weakening the consistency.

Note: nb07e (E7, arc-spectrum-subtraction sibling) was *not* refreshed because its result is a relative comparison (matches §6cum within 0.1 km/s) — a uniform per-SPS σ shift moves both arms of the comparison together and the relative match is unchanged.

H3 F200 mask sensitivity (independent) — `results/annular_bootstrap_07c_nomask/`. Same §6cum pipeline with mask off. $\sigma_e(<R_e) = 250.96$ km/s. $\Delta = -16.4$ km/s. Independently confirms the nb09 mask-weight sweep at $w=1.0$.

I. Earlier sigma-discrepancy work (`notebooks/05x`)

The 21-test diagnostic notebook resolving the $\sigma \approx 301$ km/s (190-spaxel box, NB01) vs $\sigma \approx 210$ km/s (inner circular aperture, NB05) discrepancy. Resolution: σ depends on aperture; the 190-spaxel box is contaminated by the lensed arc. This work motivated the cumulative-I path (Tests 7-13 explored masking variants, Tests 14-16 went radial, Tests 18-19 tried binning approaches). PowerBin (Test 19) was rejected. The headline analysis moved to circular apertures + cumulative I-weighting (nb06 $\rightarrow$ nb07c $\rightarrow$ nb09).

J. Wide arc-masked window (current paper headline, May 2026)

This section documents the May 8–13 work that elevated the rest 3800–5400 Å arc-masked window above the narrow Ca H+K + G-band window as the paper headline. The wide window had been off-limits in the nb07c era because of source-emission contamination from the $z=1.302$ spiral and template issues — both are now resolved.

J1 Wavelength-window sweep (nb09a) — `notebooks/09a_wavelength_window_sweep.ipynb` + `scripts/run_window_sweep.py`. Bootstrap σ_e at 15 candidate windows from def-rest 3500–5500 down to narrow Ca H+K only. Result: wR3800_5400 wins on statistical precision ($\sim 4\times$ more spectral pixels than narrow). Blue edge set by XSL template floor at def-rest 3675 Å with 5% velocity padding (J2 below); red edge set by KCWI red-channel coverage at $z=0.67564$.

Provenance of the 3800–5400 range: NOT from a specific literature prescription — it’s set by the data (XSL template floor + retaining Ca H+K to cross-anchor with the narrow w6500_7500 result). Closest literature precedent for going as blue as 3800 Å is LEGA-C (Bezanson+2018, van der Wel+2021) at $z\sim 0.7$ using ~ 3500 –5500 rest. SAURON/ATLAS3D’s canonical kinematic window is much narrower (4800–5380; Mg b + Fe5270 + Fe5335 only). The Lick-tradition “4000–5400 (no Ca H+K)” range is captured by the wR4000_5400_arcmask sensitivity check (orthogonal H β + Mg b feature set) — NOT the headline.

J2 Source-emission contamination discovery (nb09a §3a, nb09b §8) — Wide-window ppxf wild-bootstrap posterior was bimodal ($\sigma \approx 250$ alongside $\sigma \approx 100$ –150; $\text{frac}<200$ km/s = 60–80%) at $\text{deg} \geq 23$. Caused by source-frame emission lines from the $z=1.302$ spiral arc contaminating the deflector spectrum. Empirically identified as $>4\sigma$ excess above local continuum:

- def-rest 3835–3855 Å = Mg II $\lambda\lambda 2796/2803$ doublet from $z=1.302$
- def-rest 4525–4545 Å = O₂ A-band telluric leading-edge residual at obs 7593–7626 Å (NOT source emission — originally flagged as a $\sim 7.9\sigma$ wild-bootstrap excess and provisionally tagged “source rest 3300 Å feature”; relabeled 2026-05-18 after the spike was confirmed present at the same wavelength in deflector $R<1.5''$, the CLAUDE.md sky box, and a 4–8” off-deflector annulus, with deflector/off-source amplitude ratio $1.11\times$ despite a $17.5\times$ continuum-brightness ratio. See NOTES_4534A_spike_investigation_2026-05-18.md and `results/figures/spike_4534A_diagnostic.png`.)

- def-rest 5115–5135 Å = source [O II] $\lambda\lambda 3727/3729$ — sits on the Mg b absorption; was distorting the FSPS Mg b depth ratio to 1.18
- def-rest 5260–5340 Å = Mg b / [Ne III] $\lambda 3869$ cluster from source

Total: 212 of 2374 pixels (8.9% of fit window) masked.

J3 Effect of the arc mask (nb09b §8) — With arc-mask + ppxf clean=True: - σ vs polynomial degree FLAT across deg 15–29 (was bimodal at 250↔175 with a break at deg 23) - N=200 bootstrap collapses to a single sharp peak (frac<200 km/s = 0% across all 3 SPS) - clean=True drops 0 pixels at deg=21 — already clean after arc-mask - σ_e (FSPS+EMILES+XSL pool) = 254.9 –7/+5 km/s at wR3800_5400_arcmask

J4 Per-SPS template spread collapse (nb09d §1.3) —

window	FSPS	EMILES	XSL	all-3 pool	SPS spread
NARROW w6500_7500	253.7	267.9	279.6	268.0	26.0 km/s
WIDE wR3800_5400_arcmask	253.9	253.3	257.5	254.8	4.2 km/s

At the narrow window the 3 SPS libraries fundamentally disagreed by 26 km/s with strong ordering (FSPS < EMILES < XSL — see reference_sps_systematic.md). At the wide arc-masked window they agree to 4 km/s, well below per-SPS bootstrap stat error. The SPS-template uncertainty essentially disappears once ppxf has access to Mg b + Fe5270 and the broader feature set. Strong paper argument for elevating the wide window. Figure: results/figures/nb09d_per_sps_both_windows.png.

J5 Sersic2D bound-fix (scripts/run_isource_shape_sweep.py, 2026-05-11) — The I-shape sweep was inflated by a non-physical Sersic2D fit. Original script allowed $n \in [0.3, 8.0]$ and $ellip \in [0.0, 0.95]$. The F200LP fit escaped to $n=0.30$, $ellip=0.00$ — a degenerate flat-disk minimum (astropy flagged it as unsuccessful). The flat profile under-weighted the center and yielded $\sigma = 271.8$ km/s, a +20 km/s outlier inflating the I-shape std to ± 5.4 . Fix:

- Tighten bounds: $n \in [1.0, 6.0]$, $ellip \in [0.0, 0.6]$, $r_{eff} \in [r_{eff_pix} \times 0.5, r_{eff_pix} \times 2.0]$ (was $\times 0.3$ – $\times 3.0$)
- Try 3 initial-condition grids: $(n_{init}, ellip_{init}) \in \{(1.5, 0.05), (2.0, 0.2), (3.5, 0.2)\}$; keep lowest- χ^2 fit
- F200LP now converges to a physical n that gives $\sigma = 251.4$ km/s
- F140W was already converging at $n=2.14$, fix did not change it materially

After the fix, I-shape std collapsed: $\pm 5.4 \rightarrow \pm 1.5$ km/s at wide arc-masked. Universal fix applied to the I-shape sweep code path — all downstream sweeps now use these bounds.

J6 I-shape sweep at wR3800_5400_arcmask (10 shapes $\times$ 3 SPS $\times$ N=250) — results/ishape_sweep_wR3800_5400_arcmask/{shape}_{sps}_N250.npz. Ten I(r) weight maps: IFU_band, IFU_wl, F140W/F200LP $\times$ {raw, arcmasked, 1Dcog, Sersic2D}. Std across the 10 shapes, pooled across 3 SPS = 1.5 km/s. N=100 $\rightarrow$ N=250 changed result by 0.05 km/s — fully converged.

J7 F200LP spatial-mask sensitivity at wR3800_5400_arcmask (3 weights $\times$ 3 SPS $\times$ N=250) — results/maskweight_sweep_wR3800_5400_arcmask/{w00,w50,w100}_{sps}_N250.npz. w=0 (hard mask) = 254.84, w=0.5 = 250.15, w=1.0 (no spatial mask) = 247.28. Peak-to-peak / 2 = 3.8 km/s. 4 $\times$ tighter than narrow's ± 16 km/s budget because the spectral arc-emission masks now absorb the worst of the arc contamination — the spatial mask is no longer the primary defense.

J8 Three-window cross-check at N=500 (nb09 §9) — Final fit-window systematic. Three windows at full N=500:

window	rest pixels	σ_e (all-3 pool)	role
w6500_7500	obs 6500–7500 Å (rest 3879–4476)	269.7 km/s	narrow Ca H+K + G only
wR3800_5400_arcmask	rest 3800–5336	254.8 km/s	HEADLINE
wR4000_5400_arcmask	rest 4000–5400	~265 km/s	orthogonal H β + Mg b (no Ca H+K)

Spread = 15 km/s $\rightarrow$ fit-window systematic = ± 15 km/s. The wR4000_5400_arcmask window is the Lick-tradition feature-set cross-check: removes Ca H+K entirely, so any agreement between the 3800-anchored headline and 4000-anchored alternative is a check that Ca H+K is NOT artificially driving the σ_e measurement. The 10 km/s gap between WIDE and wR4000 is the dominant residual systematic.

J9 Both-windows side-by-side budget (notebooks/09d_final_systematics_both_windows.ipynb) — Cache-reuse: narrow I-shape at N=500 reused from annular_bootstrap_07c_ishape/ (only F200LP_Sersic2D and F140W_Sersic2D refit 2026-05-11 with new Sersic bounds); narrow mask-weight at N=500 reused from mask_weight_sweep.npz; both N=500 stat pools from nb09a_wavelength_sweep/. Total final budget table (also results/sigma_e_final_systematics_nb09d.npz):

component	NARROW w6500_7500	WIDE wR3800_5400_arcmask
stat (N=500)	± 23.9	± 6.1
I-shape (10 shapes, N=250, Sersic2D bound-fix applied)	± 3.7	± 1.5
F200 mask (peak-to-peak / 2)	± 7.5	± 3.8
frame (vac/air, carried)	± 5.0	± 5.0
centering (HST WCS, carried)	± 4.0	± 4.0
fit-window (3-window from §9)	± 15.0	± 15.0
TOTAL	± 30.10	± 17.87

Headline: $\sigma_e(<R_e) = 254.85 \pm 17.87$ km/s (WIDE) vs 267.95 ± 30.10 km/s (NARROW). Difference 13.1 km/s, well within both budgets at 0.4σ .

K. Resolved kinematics (per-spaxel + PowerBin at wide arc-masked window)

The wide arc-masked window unlocks resolved-kinematics measurements that were previously infeasible at the narrow window. Documented in notebooks/11_perbin_perspaxel_kinematics_wide.ipynb and memory project_nb11_resolved_kinematics.md.

K1 Per-spaxel ppxf inside $R < 1.5 R_e$ — Per-spaxel ppxf fits at S/N floors of {2, 3, 5, 10}. Results inside $R < 1.5 R_e = 3.46''$ aperture (358 spaxels total):

S/N floor	N spaxels	% σ in [150,350]	median σ	usable?
≥ 10	0	–	–	max per-spaxel S/N is 8.6
≥ 5	17	94%	201 km/s	YES — production map
≥ 3	~76	68%	~240	usable as soft map with outlier flagging
≥ 2	~134	53%	~243	mostly noise-dominated beyond $1 R_e$

$S/N \geq 5$ is the cleanest result — 17 central spaxels with $\sigma \in [144, 251]$ km/s, 0 outliers, spanning the central $\sim 1 R_e$. $\sigma(R)$ drops with radius — consistent with elliptical bulge morphology. Aperture-edge spatial resolution is fundamentally constrained by KCWI per-spaxel S/N, not ppxf or fit window.

K2 PowerBin spatial binning at wide arc-masked window — PowerBin (Cappellari 2025) at target capacity $S/N=15$. 7 bins inside $R < 1.5 R_e$; median $\sigma = 294.5$ km/s. 2 of 7 outer bins still hit $\sigma > 400$ km/s — irreducible KCWI per-spaxel S/N limit at the aperture edge. Major improvement over Test 19 narrow-window failure (which had $\sigma > 800$ outer bins) but not fully cured. Flag outer bins as noise-limited rather than attempting physical interpretation.

K3 Wide vs narrow for resolved maps — At the narrow w6500_7500 window, both per-spaxel and PowerBin attempts failed (Test 19 in nb05x — outer bins $\sigma > 800$ km/s, no per-spaxel S/N above floor). At wR3800_5400_arcmask the same machinery works because the per-pixel S/N quadruples (more line-strength signal per spaxel from the broader feature set + arc-emission masking). This is a direct qualitative gain enabled by the methodology change.

Caches & figures (nb11): - results/nb11_perbin_perspaxel_wide.npz — per-bin $\sigma/V/\chi^2$, per-spaxel $\sigma/V/\chi^2$, bin_map, V_sys - results/figures/nb11_perspaxel_sn.png — S/N map + S/N histogram - results/figures/nb11_powerbin_map.png — PowerBin bin map + per-bin S/N - results/figures/nb11_powerbin_kinematics.png — σ map, V map, $\sigma(R)$, V(R) - results/figures/nb11_perbin_perspaxel_sigma_vs_r.png — combined $\sigma(R)$

L. Paper-figure preparation (2026-05-13)

Final figure cleanup for the ApJL submission, in ../AGEL0206_ApJL_Figures/figures.ipynb.

L1 Figure 2 narrow (cell a546db7f) — Updated title to show $\sigma_e = 267 \pm 24$ (stat) ± 18 (sys) km/s (stat $\pm$ sys separately rather than a single combined ± 30). Removed the “no arc mask” blue overlay from the inset; left inset now shows a single red posterior. Saves to AGEL0206_sigma_e_SED_final.pdf.

L2 Figure 2 wide (cell fig2_wide, NEW, paper headline) — $\sigma_e = 255 \pm 6$ (stat) ± 17 (sys) km/s. I-weighted aperture spectrum over rest 3800–5400 Å with 9 absorption features labeled (Ca K, Ca H, H δ , G-band, H γ , H β , Mg b, Fe5270, Fe5335) and 4 source-emission arc masks hatched. Residual sub-panel underneath. Inset shows single red posterior (no-arc-mask comparison removed). Saves to AGEL0206_sigma_e_SED_final_wide.pdf.

L3 Figure 3 M_{BH}–M_{star} cleanup (cell fig3_clean_no_err, NEW) — Cleaned version of the comparison plot. NO per-object error bars, uniform medium-gray color for all local early-types (no E vs S0 / KH13 vs post-KH13 subdivision). Filtered to Greene+2020 E+S0 list (60 unique galaxies): dropped 19 post-KH13 additions that were not in Greene+2020 Fig 10 (mostly Saglia+2016 / Thater+2019 fills), matched 4 KH13 suffix variants (NGC 1316b / 2778c / 3998c / 4486A) to their Greene+2020 names (NGC 1316 / 2778 / 3998 / 4486A). All literature relations (Greene+20, Reines & Volonteri+15, Farrah+25, Sahu+24) preserved. AGEL0206 point at $M_\star=10^{11.3}$, M_{BH}=6.5e8 preserved. Saves to Mbh_Mstar_relation_clean.pdf. The legacy cell 33 with per-object error bars and E vs S0 color subdivision is preserved as Mbh_Mstar_relation.pdf.

Sanity / withdrawn: during the wide-window writeup an erroneous “Cappellari 2017 recommends extending blueward” citation was drafted in chat — retracted, no such recommendation exists in the ppxf paper. The wavelength-range choice is data-driven (XSL template floor + retain Ca H+K), with LEGA-C (van der Wel+21) as the closest literature precedent (see J1 above).

3. File index

Documents (top-level)

File	Purpose
TESTS_AND_DIAGNOSTICS.md	This file — master test index
NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md	Frame fix derivation + addenda (a)–(d)
HANDOVER.md	Apr 27 σ_e analysis snapshot (pre-frame-fix; superseded by nb09)
HANDOFF_Ir_sweep_2026-04-28.md	I(r) sweep design doc (consumed; sweep complete)
NOTES_methodology_2026-04-27.md	Cube provenance + earlier methodology
NOTES_centering_investigation_2026-04-27.md	5-center sweep result + ± 4 km/s budget
CLAUDE.md	Repo conventions, key results summary
README.md	High-level project description

Notebooks (paper-ready / current)

Notebook	Purpose
notebooks/09_final_sigma_e_paper.ipynb	Paper-ready σ_e at both windows; §9 three-window cross-check, §10 I-shape N=250, §10.1 mask-weight N=250
notebooks/09a_wavelength_window_sweep.ipynb	15-window scan; identifies wR3800_5400 sweet spot

Notebook	Purpose
<code>notebooks/09b_lsf_and_sigma_clip.ipynb</code>	Arc-mask + ppxf clean=True diagnostic with 3×3 zoom plots
<code>notebooks/09d_final_systematics_both_windows.ipynb</code>	Side-by-side WIDE vs NARROW final budget (paper headline writeup)
<code>notebooks/11_perbin_perspaxel_kinematics_wide.ipynb</code>	Per-spaxel + PowerBin σ map at wR3800_5400_arcmask (NEW — previously infeasible)
<code>notebooks/04_redshift_verification.ipynb</code>	$z = 0.67564$ line-fit
<code>notebooks/02_streamlined_Bagpipes_SED.ipynb</code>	$\log M_{\star} = 11.33$
<code>notebooks/08_sersic_total_photometry.ipynb</code>	Sersic-total photometry cross-check
<code>notebooks/07c_sigma_e_equalN.ipynb</code>	§6cum + §7 equal-N annular cross-check (narrow window, retained for cross-check)
<code>notebooks/07e_sigma_e_arc_subtract.ipynb</code>	Arc-subtraction sibling cross-check
<code>notebooks/05x_sigma_discrepancy_diagnostic.ipynb</code>	Tests 1-21 (foundational diagnostics)

Notebooks (history / reference, not paper-driving)

01_*, 02_*, 03_bootstrap_ppxf_errors, 03b_*, 03c_*, 05_*, 06_*, 07_*, 07a_*, 07b_*, 07d_* — kept for the history of the analysis.

Scripts (production)

Script	Purpose
<code>scripts/final_sigma_e.py</code>	nb09 production driver — runs all 3 mask tracks at N=500
<code>scripts/build_nb09.py</code>	Generates the paper notebook from the saved artifacts
<code>scripts/bootstrap_ppxf.py</code>	Frame-aware ppxf input prep + wild bootstrap
<code>scripts/bootstrap_ppxf_parallel.py</code>	joblib parallel bootstrap (BLAS=1 per worker)
<code>scripts/measure_Re.py</code>	R_e via masked CoG (multi-band, multi-strategy)
<code>scripts/redshift_verify.py</code>	Line-fitting redshift module
<code>scripts/ifu_spectral_resolution.py</code>	LSF audit

Scripts (sweeps and audits)

Script	Purpose
<code>scripts/audit_ppxf_methodology.py</code>	4 orthogonal correctness audits
<code>scripts/sigma_inst_sensitivity.py</code>	DISPSCAL × {0.5..2.0} sweep
<code>scripts/run_isource_shape_sweep.py</code>	10-shape $I(r)$ sweep — Sersic2D bound-fix applied 2026-05-11 ($n \in [1.0, 6.0]$ + multi-init grid)
<code>scripts/analyze_isource_shape_sweep.py</code>	I-shape sweep analysis + figure
<code>scripts/run_window_sweep.py</code>	Wavelength-window driver with arc-mask flag; underpins nb09a + nb09 §9
<code>scripts/soft_mask_track.py</code>	Mask-weight runs (CLI <code>--weight</code>)
<code>scripts/analyze_mask_weight_sweep.py</code>	5-point sweep analysis + linearity fit
<code>scripts/regen_s6cum_nomask_diagnostic.py</code>	Centering investigation
<code>scripts/relock_nomask_Re_N500.py</code>	One-off N=500 relock for the no-mask track

Result files (key)

File	Contents
WIDE arc-masked (current headline)	
<code>results/sigma_e_final_systematics_nb09d.npz</code>	Both-windows side-by-side budget (paper headline writeup)
<code>results/nb09a_wavelength_sweep/wR3800_5400_arcmask_*_T*_N500.npz</code>	N=500 stat pool at headline window

File	Contents
results/nb09a_wavelength_sweep/wR4000_5400_arcmask_*_T*_N500.npz	N=500 orthogonal-feature-set cross-check
results/ishape_sweep_wR3800_5400_arcmask/	10 shapes $\times$ 3 SPS $\times$ N=250 (I-shape ± 1.5)
results/maskweight_sweep_wR3800_5400_arcmask/	3 weights $\times$ 3 SPS $\times$ N=250 (mask ± 3.8)
results/sigma_e_window_sweep_09a.npz	15-window posterior summary
results/nb11_perbin_perspaxel_wide.npz	Per-bin + per-spaxel kinematics at wide
NARROW Ca H+K + G (cross-check)	
results/final_sigma_e_paper.npz	nb09 narrow headline + per-SPS + 3-track summaries
results/final_sigma_e_paper/	Per-(aperture, SPS, mask_weight) caches
results/sigma_e_radial_07c.npz	nb07c §6cum + §7 + §7b posteriors
results/annular_bootstrap_07c/	Per-aperture cumulative I-weighted caches
results/annular_bootstrap_07c_ishape/	Per-(shape, SPS) I-shape caches (N=500, F140W/F200LP Sersic2D refit 2026-05-11)
results/annular_bootstrap_07c_nomask/	§6cum nomask caches
results/mask_weight_sweep.npz	5-point mask-weight sweep summary (narrow)
Audits & figures	
results/ppxf_methodology_audit.npz	4-audit results
results/figures/nb09_*.png	nb09 paper figures (incl. fit windows + I-shape comparison)
results/figures/nb09d_per_sps_both_windows.png	Per-SPS posterior collapse (26 $\rightarrow$ 4 km/s)
results/figures/nb11_*.png	Resolved-kinematics maps (per-spaxel S/N, σ map, V map, $\sigma(R)$, V(R))
results/figures/nb07c_ishape_sweep.png	I-shape sweep figure
../AGEL_0206_ApJL_Figures/AGEL0206_sigma_e_SED_final_wide.pdf	Paper Figure 2 (wide, headline)
../AGEL_0206_ApJL_Figures/AGEL0206_sigma_e_SED_final.pdf	Paper Figure 2 (narrow, cross-check)
../AGEL_0206_ApJL_Figures/Mbh_Mstar_relation_clean.pdf	Paper Figure 3 (clean, no error bars, Greene+2020-matched)

4. Reproduction recipes

conda activate ISMgas

```
# === Headline pipeline (~50 min at N=500) ===
python scripts/final_sigma_e.py --n_bootstrap 500
python scripts/build_nb09.py
jupyter nbconvert --to notebook --execute --inplace notebooks/09_final_sigma_e_paper.ipynb
```

```
# === Methodology audits (~3 min) ===
python scripts/audit_ppxf_methodology.py
python scripts/sigma_inst_sensitivity.py
```

```
# === I-shape sweep (~50 min at N=500) ===
python scripts/run_isource_shape_sweep.py --n_bootstrap 500 --n_jobs 8
python scripts/analyze_isource_shape_sweep.py
```

```
# === Mask-weight sweep (~25 min total) ===
# (w=0 and w=1 are produced by final_sigma_e.py; only need 0.25/0.5/0.75)
python scripts/soft_mask_track.py --weight 0.25
python scripts/soft_mask_track.py --weight 0.50
python scripts/soft_mask_track.py --weight 0.75
python scripts/analyze_mask_weight_sweep.py
```

```
# === Smoke run (~6 min, useful for rebuilds) ===
```

```
python scripts/final_sigma_e.py --n_bootstrap 50
```

Caches are keyed by N_bootstrap and mask_weight in the filename — re-runs at the same N skip cached fits unless --force is passed.

5. What's intentionally NOT done

Updated 2026-05-13 with status changes after the wide arc-masked window work and the Sersic2D bound-fix.

- Re-run nb09 at higher N (e.g. N=2000): the pooled posterior is already smooth; at the wide arc-masked window the stat error is ± 6 km/s while the fit-window systematic dominates at ± 15 km/s. More N is wasted compute.
- ~~Re-fit Sersic2D-F200LP~~ DONE 2026-05-11 — the $n=0.30$ fit was a parameter-bound pathology, not a UV-arc artifact. Bound-fix (J7) tightened bounds to $n \in [1.0, 6.0]$ with multi-init grid. F200LP_Sersic2D now converges to a physical fit ($\sigma=251.4$ km/s) and is kept in the I-shape budget, not flagged as an outlier.
- Voronoi binning — Test 18 / nb05x — abandoned; not pursued at N=500.
- ~~PowerBin spatial binning~~ PARTIAL — Test 19 narrow-window rejected was correct at the time. At the wide arc-masked window (K2 / nb11) PowerBin now works for the inner ~ 5 of 7 bins; 2 outer bins still hit $\sigma > 400$ (irreducible KCWI per-spaxel S/N at aperture edge). Not used as headline binning scheme but published as supplementary σ map.
- ~~Per-spaxel rotation map at production statistics~~ DONE (K1 / nb11) — at the wide arc-masked window, $S/N \geq 5$ gives 17 central spaxels with clean σ values and a visible $\sigma(R)$ gradient. No coherent rotation detected above the noise (consistent with elliptical-bulge expectation). Published as supplementary kinematic map.
- Full nb07a Sersic-I revisit at the post-frame-fix headline — superseded by the I-shape sweep (D3/D4/J6) which covers the Sersic2D shape with the bound-fix applied.
- Re-quote σ_e on KH13/Bell+2003 stellar masses for the M_BH-M★ plot — Figure 3 keeps the KH13 dual-M/L dynamical bulge masses for the local comparison sample. The Greene+2020 list match (L3) restricts to the same 60 galaxies as Greene+2020 Fig 10 but with our (better) bulge mass values.
- Cross-match nb09a “narrow” window to other AGEL targets — paper is single-target. The arc-emission-mask methodology (J2/J3) is reusable for future AGEL targets where source z is known, but applying it elsewhere is not in scope for the ApJL.

See `/claude/projects/.../memory/MEMORY.md` for compressed cross-session references, including: - `reference_cumulative_vs_annular_sigma_e.md` — §6cum vs §7 design choice - `reference_sps_systematic.md` — per-SPS V_{sys} + frame-fix details - `reference_ppxf_vacair_handling.md` — vac/air conversion methodology - `reference_kcwi_data_properties.md` — KCWI cube + LSF + paper boilerplate - `project_nb09_final_sigma_e.md` — current headline status